

nnActive

A Framework for Evaluation of Active Learning in 3D Biomedical Segmentation

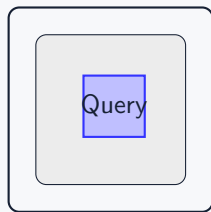
Carsten T. Lüth et al.

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Does active learning save annotation effort?

Why Active Learning for 3D Biomedical Segmentation?

- 3D biomedical segmentation needs accurate voxel-level labels.
- Expert annotation is expensive and often requires specialist knowledge.
- Active Learning asks: **which region should we annotate next?**
- The goal is not just higher Dice Score, but **less annotation effort**.



3D scan / patch

Annotate only the most useful regions

Why Is Better Evaluation Needed?

- Prior work gives **mixed conclusions** on whether Active Learning really beats Random sampling.
- A useful AL method should work **across datasets and annotation budgets**.
- It should still help when combined with other ways of using annotations efficiently (self- or semi-supervised learning).
- It should beat **realistic and inexpensive baselines**, not only naive Random sampling.
- Evaluation should measure realistically **annotation effort**, not only model performance.

Core question

Can we trust reported Active Learning gains in realistic 3D biomedical settings?

Four Pitfalls in Active Learning Evaluation

P1: Too Few Settings

Evaluation on too few datasets and annotation budgets, so result may not generalise.

P3: Weak Random Baseline

Uniform Random often selects mostly background, making Active Learning may look artificially strong

P2: Partial Annotations Used Poorly

2D training discards useful surrounding 3D context.

P4: Voxels $\neq$ Annotation Effort

Equal-size patches can require very different annotation effort.

How nnActive Addresses the Pitfalls

P1 → Broader Evaluation

4 biomedical datasets and 3 annotation-budget regimes.

P3 → Stronger Baselines

Introduces Foreground-Aware Random sampling.

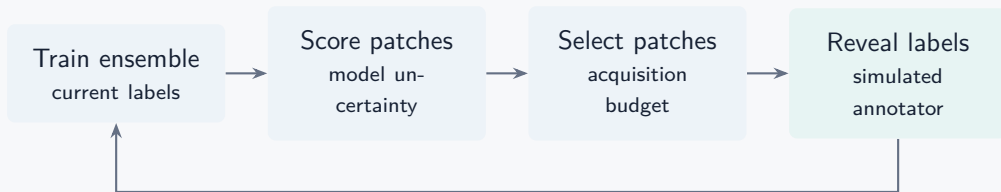
P2 → Realistic 3D Training

3D nnU-Net trained with partial annotations and surrounding context.

P4 → Better Cost Metric

Introduces Foreground Efficiency (FG-Eff) as a proxy for annotation effort.

The acquisition loop



Strong segmentation backbone

Five 3D full-resolution nnU-Net models, trained via five-fold cross-validation.

200 epochs per model; retrain from scratch.

Shared acquisition protocol

Voxel scores averaged over query patches.

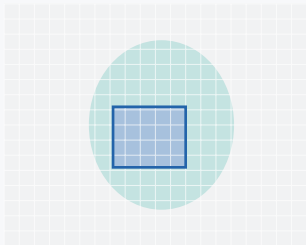
No overlap with previously selected patches.

Starting labels cover every foreground class.

nnActive, §4–5.1. Ground-truth labels are revealed to simulate annotation.

A small annotation can use a larger context

model input: surrounding image



query patch: labeled region

Image context is available everywhere.

Supervision is available only on the labeled voxel set S :

$$\mathcal{L}(\theta) = \sum_{v \in S} \ell(f_{\theta}(X)_v, Y_v).$$

Unlabeled voxels are ignored by the loss.

Region sampling varies the labeled voxel's position inside the model input.

Query size and model input size are different.

nnActive, Appendix C.3. Equation illustrates the supervision mask, not the full nnU-Net loss.

Uncertain predictions or disagreement?

For M models, $\bar{p}(c | x) = \frac{1}{M} \sum_m p_m(c | x)$.

Predictive Entropy

$$H(\bar{p}) = - \sum_c \bar{p}_c \log \bar{p}_c$$

Select patches with uncertain average predictions.

BALD

$$H(\bar{p}) - \frac{1}{M} \sum_m H(p_m)$$

Select patches where ensemble members disagree.

Binary example, same mean prediction	Entropy	BALD
Every model predicts (0.5, 0.5)	High	Zero
Half predict (1, 0), half (0, 1)	High	High

nnActive, §4 and Appendix C. Ensemble BALD approximates Bayesian model disagreement.

Why add noise to patch selection?

Greedy selection

Take the highest uncertainty scores.

- Predictive Entropy
- BALD

Risk: many selected patches may look alike, even without spatial overlap.

Noisy selection

Perturb scores or ranks before selecting.

- PowerPE
- PowerBALD
- SoftrankBALD

Randomness can broaden selection, especially when labels are scarce.

Highest uncertainty does not guarantee the best batch.

nnActive, §4 and §5.2. Randomized selection does not guarantee semantic diversity. Main study: $\beta = 1$.

A benchmark across tasks and budgets

Dataset	Segmentation targets	Low / medium / high
ACDC	Three cardiac structures, MRI	150 / 300 / 450
AMOS2022	Fifteen organs	200 / 1,000 / 2,500
KiTS2021	Kidney, tumor and cyst	200 / 1,000 / 2,500
Hippocampus	Anterior and posterior regions	100 / 200 / 300

Budgets are numbers of query patches; patch dimensions depend on the dataset.

Matched comparisons

75% training/acquisition pool; 25% test.

Eight methods; four acquisition seeds.

Five budget points

20%, 40%, 60%, 80%, 100%.

Shared initial labels; four further acquisitions.

nnActive, §5.1 and Appendix E. More than 7,500 model trainings including ablations.

Foreground-aware random sampling

Uniform Random

Choose patches anywhere in the volume.

Many queries can land on easy background.

Random 66% FG

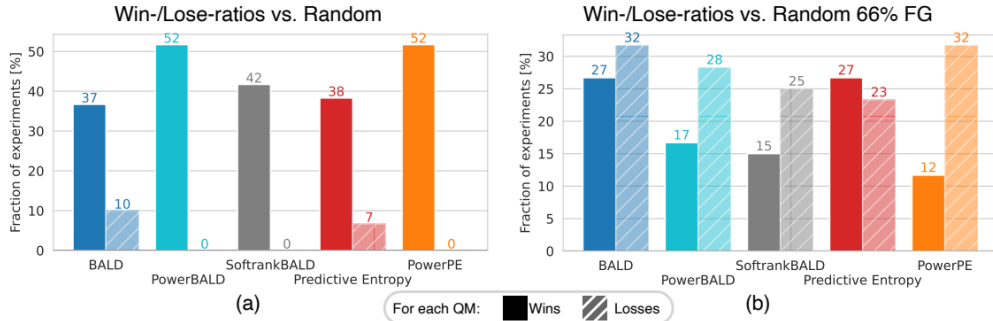
Approximately one-third of queries are uniform.
Two-thirds target foreground:

- Half centered on a sampled foreground class.
- Half centered on a class boundary.

**Can uncertainty selection beat cheap,
anatomy-aware sampling?**

nnActive, §4. Random 33% FG uses the same mechanism less often. Ground truth simulates localization; real screening cost is unmeasured.

The baseline changes the verdict



Against uniform Random: many more wins than losses.

Against Random 66% FG: only Predictive Entropy has a positive win/loss balance.

nnActive, Figure 3. Bars show significant pairwise wins/losses across comparisons (t -tests, $\alpha = .05$), not Dice gains.

Results for different datasets

Final Dice $\times 100$	Random	Random 66% FG	Entropy
AMOS, low budget	36.36	71.11	39.19
KiTS, medium budget	48.41	51.67	65.39

AMOS: cover all organs

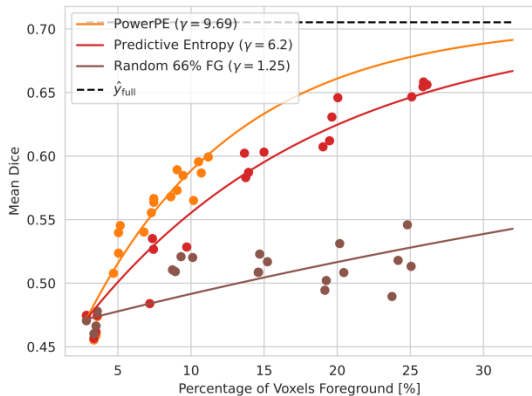
Foreground-aware sampling covers small structures. Greedy uncertainty can focus repeatedly on a subset of classes.

KiTS: learn difficult background

Uncertainty selection can target background regions causing false positives. Foreground sampling can miss them.

nnActive, Table 6 and §5.2 Q4. Compare within rows. Values are reported means; explanations are the authors' interpretations.

Measuring annotation effort



Foreground Efficiency

Fit how quickly the gap to fully labeled performance closes as annotated foreground increases. Larger γ means faster closure.

KiTS, medium budget

PowerPE: $\gamma = 9.69$, Dice 58.67.

Entropy: $\gamma = 6.20$, Dice 65.39.

Better efficiency can coexist with lower final accuracy.

nnActive, Figure 7 and Table 6. Foreground count is a proxy for human effort. Compare γ only within matched setups.

What changes the acquisition outcome?

Ablation	Finding and trade-off
Smaller acquisition batches	More frequent feedback generally helps, but requires more re-training and inference.
Longer training	Better-trained models also select better patches; the gain is not only better final fitting.
More selection noise	Often helps at low budgets; less noise can help later. No universal best setting.
Smaller query patches	Relative method rankings change. Halving each dimension labels eight times fewer voxels per patch.

nnActive, §5.3 and Appendix G. Smaller patches do not establish equal accuracy with eight times less annotation.

Conclusion

Strong evidence

- Strong 3D segmentation pipeline.
- Diverse tasks, budgets and seeds.
- Stronger baselines and cost proxies.

Remaining assumptions

- Foreground localization is simulated with ground truth.
- Foreground voxels do not measure annotation minutes.
- Five uncertainty methods do not cover all active learning.

**Structured starting labels already cover every class.
The study does not test a fully unlabeled cold start.**